# Enterprise ERP, CRM & POS WhatsApp Integration Manual

Complete Technical Specification for Connecting Repair Billing, Order Fulfillment & Commerce Systems to Meta WhatsApp Cloud API

| | |
|---|---|
| **Production Base URL:** | **https://whatapp-automation-kxml.onrender.com/api/v1** |
| **Interactive Hub:** | **https://itechmobileswhatsappautomation.vercel.app** |
| **API Authentication:** | Header: x-api-key: itech_live_sec_7f9984e1b9384e9d9a1f |
| **Meta WABA Account ID:** | 1758570912097342 (Official Verified Business Account) |
| **Store Helpline & Tech:** | +91 9361863063 · email: itechnicmobiles2@gmail.com · Tirukannur, Pondicherry |

## 1. System Architecture & Delivery Flow

Your billing or CRM software dispatches standard HTTPS REST requests to the gateway whenever repair cards or orders update. The gateway automatically maps each event to pre-approved Meta Utility Templates, ensuring 24/7 delivery outside WhatsApp's 24-hour window.

## 2. Authentication & Quick-Start Connectivity Test

Send x-api-key: itech_live_sec_7f9984e1b9384e9d9a1f in HTTP headers with all requests:

```
GET /api/v1/auth/key HTTP/1.1
Host: whatapp-automation-kxml.onrender.com
x-api-key: itech_live_sec_7f9984e1b9384e9d9a1f

// Returns 200 OK: {"success": true, "apiKey": "itech_live_sec...", "wabaId": "1758570912097342"}
```

## 3. CRM Service Desk & Repair Lifecycle Endpoints

Trigger these 6 endpoints from your repair job card module during device servicing:

| Endpoint & Method | ERP Trigger Event | Key JSON Payload Parameters |
|---|---|---|
| **POST** /crm/repair-booking-confirmed | Customer creates repair job card | phone, customerName, jobId, device, service, turnaround |
| **POST** /crm/repair-status-update | Technician updates status (Diagnosing, QA) | phone, customerName, jobId, device, status |
| **POST** /crm/device-ready-pickup | Device passes QA and invoice is generated | phone, customerName, jobId, device, finalCost, billNo |
| **POST** /crm/service-quotation | Diagnostic cost approval needed from customer | phone, customerName, jobId, device, issue, quotedCost |
| **POST** /crm/repair/send-warranty | Payment settled; sends 90-day parts warranty | phone, repairId |
| **POST** /crm/doorstep-pickup | Customer books technician doorstep visit | phone, customerName, timeSlot, address |

### 3.1 Service Desk JSON Payloads & Delivery Previews

Exact JSON payloads to transmit from your billing/repair software:

```
// 1. Repair Booking Confirmed -> POST /api/v1/crm/repair-booking-confirmed
{
  "phone": "919876543210",
  "customerName": "Ramesh Babu",
  "jobId": "REP-2026-081",
  "device": "Samsung Galaxy S23",
  "service": "Original OLED Screen Replacement",
  "turnaround": "Same Day (60 Mins)"
}

// 2. Service Quotation Approval -> POST /api/v1/crm/service-quotation
{
  "phone": "919876543210",
  "customerName": "Ramesh Babu",
  "jobId": "REP-2026-081",
  "device": "Samsung Galaxy S23",
  "issue": "Cracked OLED Display",
  "quotedCost": "4500"
}

// 3. Device Ready for Pickup & Invoice -> POST /api/v1/crm/device-ready-pickup
{
  "phone": "919876543210",
  "customerName": "Ramesh Babu",
  "jobId": "REP-2026-081",
  "device": "Samsung Galaxy S23",
  "finalCost": "4500",
  "billNo": "INV-89104"
}
```

## 4. Retail Commerce, Orders & Warehouse Synchronization

Sync products, warehouse inventory, order milestones, and hot sales leads:

| Endpoint & Method | Capability & Purpose | Parameters / Usage |
|---|---|---|
| **PUT** /orders/:id/status | Update milestone: Confirmed -> Processing -> Shipped -> Out for Delivery -> Delivered | orderStatus, courierPartner, trackingNumber, assignedRider: {name, phone} |
| **POST** /orders/:id/send-update | Dispatches live 5-stage milestone card to customer WhatsApp | URL param :id (e.g. ORD-1082). Call after status change. |
| **PUT** /products/:id | Warehouse stock & price sync (ERP -> WhatsApp Bot) | stock, price, offerPrice, inStock |
| **GET** /leads | Fetch high-intent phone buyer leads captured by WhatsApp Bot | Returns lead list with budget, target device, phone, and purchase intent. |
| **PUT** /leads/:id/status | Update lead status in CRM (New, Contacted, Qualified, Closed) | status, notes |
| **POST** /leads/:id/send-whatsapp | Sales executive sends direct WhatsApp follow-up from CRM | messageText (Logged to customer chat ledger) |

```
// 4. Order Milestone WhatsApp Dispatch -> POST /api/v1/orders/ORD-1082/send-update
{
  "orderStatus": "Out for Delivery",
  "courierPartner": "iTechnic Express Rider",
  "trackingNumber": "TRK-99214",
  "assignedRider": { "name": "Karthik Raja", "phone": "919361863063" }
}

// 5. Warehouse Stock & Price Sync -> PUT /api/v1/products/PROD-001
{
  "stock": 14,
  "price": 69999,
  "offerPrice": 64999,
  "inStock": true
}
```

## 5. Step-by-Step CRM & ERP Integration Blueprints

Architectural blueprints for connecting different software architectures:

| Architecture Pattern | How to Implement | Recommended Trigger Points |
|---|---|---|
| **Custom Web CRM / ERP**<br>PHP, Node, Python, Java | Add an HTTP client helper class. Call WhatsApp endpoints on backend database commit hooks. | - On Job Card creation<br>- On status dropdown change<br>- On invoice generation |
| **Odoo / ERPNext**<br>Python Automated Actions | Create an Automated Action / Server Action using Python requests to POST payload on state transitions. | - action_confirm()<br>- button_repair_ready()<br>- action_invoice_paid() |
| **Desktop POS / Billing**<br>C# .NET, WPF, Java, VB | Use HttpClient to dispatch async POST request in a background thread when printing invoice. | - On Print Bill button click<br>- On Job Card status save |
| **No-Code / Cloud CRM**<br>Zoho CRM, Salesforce, Zapier | Configure standard Outgoing Webhook rule. Map lead/ticket fields to JSON payload with x-api-key. | - Workflow Rule: When Stage is 'Quote Sent' or 'Closed Won' |

## 6. Multi-Language Code Snippets for Developers

Copy and paste these production helper functions directly into your backend:

### 6.1 PHP (Laravel, CodeIgniter, WordPress, Custom PHP)

```php
<?php
class WhatsAppService {
    private static $baseUrl = 'https://whatapp-automation-kxml.onrender.com/api/v1';
    private static $apiKey  = 'itech_live_sec_7f9984e1b9384e9d9a1f';

    public static function send($endpoint, $payload) {
        $ch = curl_init(self::$baseUrl . $endpoint);
        curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
        curl_setopt($ch, CURLOPT_POST, true);
        curl_setopt($ch, CURLOPT_POSTFIELDS, json_encode($payload));
        curl_setopt($ch, CURLOPT_HTTPHEADER, [
            'Content-Type: application/json',
            'x-api-key: ' . self::$apiKey
        ]);
        $res = curl_exec($ch);
        curl_close($ch);
        return json_decode($res, true);
    }
}

// Example: Call when repair job is completed:
WhatsAppService::send('/crm/device-ready-pickup', [
    'phone' => '919876543210', 'customerName' => 'Ramesh', 'jobId' => 'REP-1042',
    'device' => 'iPhone 13', 'finalCost' => '4500', 'billNo' => 'INV-2026-908'
]);
```

### 6.2 Node.js / TypeScript (Express, NestJS, Next.js)

```javascript
const axios = require('axios');
const wa = axios.create({
  baseURL: 'https://whatapp-automation-kxml.onrender.com/api/v1',
  headers: { 'x-api-key': 'itech_live_sec_7f9984e1b9384e9d9a1f', 'Content-Type': 'application/json' }
});

async function notifyRepairBooking(phone, customerName, jobId, device, service) {
  return await wa.post('/crm/repair-booking-confirmed', { phone, customerName, jobId, device, service, turnaround: 'Same Day' });
}

async function notifyOrderMilestone(orderId, orderStatus, courierPartner, trackingNumber) {
  return await wa.post(`/orders/${orderId}/send-update`, { orderStatus, courierPartner, trackingNumber });
}
```

### 6.3 Python (Odoo, ERPNext, Django, FastAPI)

```
import requests
BASE_URL = 'https://whatapp-automation-kxml.onrender.com/api/v1'
HEADERS  = {'x-api-key': 'itech_live_sec_7f9984e1b9384e9d9a1f', 'Content-Type': 'application/json'}

def notify_quote(phone, customer_name, job_id, device, issue, quoted_cost):
    payload = {'phone': phone, 'customerName': customer_name, 'jobId': job_id, 'device': device, 'issue': issue, 'quotedCost': str(q
    return requests.post(f'{BASE_URL}/crm/service-quotation', json=payload, headers=HEADERS, timeout=10).json()

def sync_product_stock(product_id, stock_count, price, offer_price):
    payload = {'stock': stock_count, 'price': price, 'offerPrice': offer_price, 'inStock': stock_count > 0}
    return requests.put(f'{BASE_URL}/products/{product_id}', json=payload, headers=HEADERS, timeout=10).json()
```

### 6.4 C# / .NET (Desktop POS, WPF, Windows Forms)

```
using System.Net.Http;
using System.Text;
using System.Text.Json;

public class WhatsAppIntegration {
    private static readonly HttpClient client = new HttpClient { BaseAddress = new Uri("https://whatapp-automation-kxml.onrender.com
    static WhatsAppIntegration() { client.DefaultRequestHeaders.Add("x-api-key", "itech_live_sec_7f9984e1b9384e9d9a1f"); }

    public static async Task SendPickupReady(string phone, string name, string jobId, string device, string cost, string billNo) {
        var data = new { phone, customerName = name, jobId, device, finalCost = cost, billNo };
        var content = new StringContent(JsonSerializer.Serialize(data), Encoding.UTF8, "application/json");
        await client.PostAsync("crm/device-ready-pickup", content);
    }
}
```

## 7. Critical Developer Rules, Rate Limits & Compliance

**CRITICAL DEVELOPER GUIDELINES:**

- **24-Hour Customer Window Bypass:** Regular WhatsApp session messages expire after 24 hours of customer inactivity. All endpoints in this guide map to **pre-approved Meta Utility Templates**. They bypass this window and deliver **24/7/365**.
- **Phone Number Normalization:** Pass mobile numbers as strings with country code (e.g. 919876543210). If a 10-digit number is provided, the gateway automatically normalizes it by prepending 91.
- **Two-Way Webhooks:** Customer replies and delivery receipts (Sent, Delivered, Read) are ingested at /webhook and synchronized to the centralized customer ledger in MongoDB.
- **Interactive Developer Portal:** Test every endpoint live and view test responses directly at: **https://itechmobileswhatsappautomation.vercel.app** (under the *API Hub* tab).
- **Retry & Idempotency Policy:** In the event of 5xx errors or network timeouts, retry up to 3 times with exponential backoff (5s, 15s, 30s). Unique Job IDs prevent duplicate notifications.

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